

Static models with elastic demand

Monday May 04, 2026

400-level Transportation systems courses

- 405: Asphalt Materials I
- 406: Pavement Design I
- 407: Airport Design
- 409: Railroad Track Engineering
- 411: Railroad Project Design and Construction
- 412: High-speed Rail Engineering
- 415: Geometric Design of Roadways
- 416: Traffic Capacity Analysis
- 417: Transportation Planning
- 418: Public Transportation Systems
- 419: Transportation Economics

Static economic models

- Consider a *single* route between two points
- Unlike in the network equilibrium models. . .
 - There is no *choice* of route
 - The overall demand is **endogenous** (i.e., it can vary)

Static economic models

- Let t (tu) be the travel time on the road
- Let q (veh/tu) be the flow
- Let p (\$/veh) be the *toll*
- $t = T(q)$ (veh/tu) is an increasing *supply function* (such as the BPR function)
- $q = D(t, p)$ (veh/tu) is *demand function* that decreases with the travel time and toll (i.e., the longer a trip takes and the higher the toll, the fewer people want to drive)
- Given a toll p , what are the **equilibrium** travel time and flow: t^e and q^e ?

Equilibrium

- Given a toll p , the **equilibrium** is a flow/travel time combination (q^e, t^e) that satisfies:

$$q^e = D(t^e, p) \quad \text{and} \quad t^e = T(q^e) \quad (1)$$

- The travel time t^e invites a flow $q^e = D(t^e, p)$ (via the demand function), which causes the travel time to be $t^e = T(q^e)$ (via the supply function)
- Or you could just solve in terms of q^e or t^e directly:

$$t^e = T(D(t^e, p)) \qquad q^e = D(T(q^e), p) \quad (2)$$

Numerical example 1

Numerical Example 1

- Suppose $D(t, p) = 100 - p - 10t$ and $T(q) = 1 + 0.001q^2$
- Suppose $p = 5$
- What are the equilibrium flow and travel time?

Numerical example 1

- Suppose $D(t, p) = 100 - p - 10t$ and $T(q) = 1 + 0.001q^2$
- What are the equilibrium flow and travel time when $p = 5$?

$$q^e = D(t^e, 5) \qquad t^e = T(q^e) \qquad (3)$$

$$q^e = 100 - 5 - 10t^e \qquad t^e = 1 + 0.001(q^e)^2 \qquad (4)$$

$$q^e = 95 - 10 \cdot (1 + 0.001(q^e)^2) \qquad (5)$$

$$0 = -.01(q^e)^2 - q^e + 85 \qquad (6)$$

- Solve for q^e with quadratic formula:

$$q^e = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad (7)$$

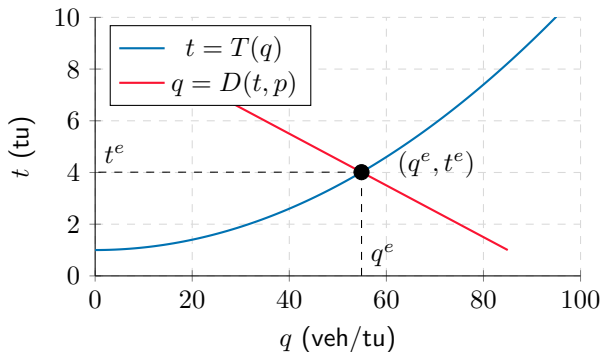
$$q^e = \frac{1 \pm \sqrt{1 + 4 \cdot 0.01 \cdot 85}}{-2 \cdot 0.01} \qquad (8)$$

$$q^e = -154.9 \quad \text{or} \quad q^e = 54.9 \qquad (9)$$

- Flow is $q^e = 54.9$.
- Travel time is $t^e = T(q^e) = 1 + 0.001(54.9)^2 = 4.01$.

Equilibrium depicted visually

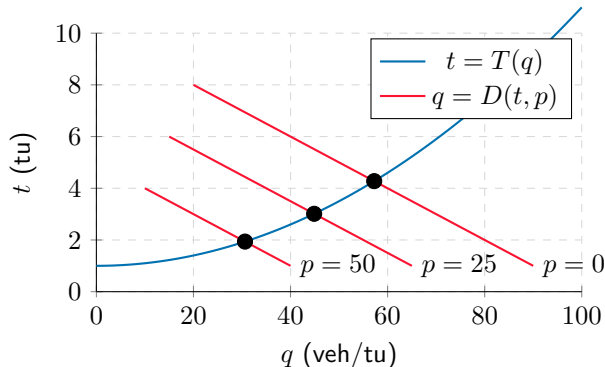
- A familiar way to draw this is to use an *inverse demand function*



Tolling

Tolling

- As tolls rise, what happens to the equilibrium flow and travel time?
- Here I plot $D(t, p) = 100 - p - 10t$ and $T(q) = 1 + 0.001q^2$ for $p = 0, 25, 50$:



- You can see that the toll lowers the flow and travel time.

Numerical example 2

Numerical example 2

- Suppose $D(t, p) = 100 - p - 10t$ and $T(q) = 1 + 0.001q^2$
- What toll would achieve an equilibrium travel time of $t^e = 3$?
- Before, we wrote the equilibrium conditions as: $q^e = D(t^e, p)$ and $t^e = T(q^e)$
- We are looking for a p such that $t^e = 3$. Use our definition of equilibrium:

$$q^e = D(t^e, p) \qquad t^e = T(q^e) \qquad (10)$$

$$q^e = 100 - p - 10t^e \qquad t^e = 1 + 0.001(q^e)^2 \qquad (11)$$

$$q^e = 100 - p - 10(3) \qquad 3 = 1 + 0.001(q^e)^2 \qquad (12)$$

- Substitute q^e and solve for p :

$$3 = 1 + 0.001 \cdot \{70 - p\}^2 \qquad (13)$$

$$p = 25.28 \qquad (14)$$

- The equilibrium flow is $q^e = 100 - 10 \cdot 3 - 25.28 = 44.72$.
- For $p = 5$, the equilibrium flow was $q^e = 54.9$. So flow fell 18.5% but travel time fell 25%.

Induced demand

Induced Demand

- **Induced demand** means that when road capacity increases, generalized travel cost falls and more trips are taken
- In this static model, a capacity increase shifts the supply curve $t = T(q)$ downward/rightward
- With demand held fixed, the new equilibrium typically has:
 - higher equilibrium flow (q^e shifts outward)
 - lower travel time than before, but often not as low as the capacity gain alone would suggest
- Intuition: some of the capacity benefit is “used up” by additional traffic

Induced Demand: Numerical Example

- Keep inverse demand fixed at $t = 8 - 0.08q$
- Before expansion: $T_0(q) = 1 + 0.05q$
- After expansion (higher capacity): $T_1(q) = 0.5 + 0.03q$
- Equilibrium before:

$$1 + 0.05q = 8 - 0.08q \quad (15)$$

$$q_0^e = 53.85, \quad t_0^e = 3.69 \quad (16)$$

- Equilibrium after:

$$0.5 + 0.03q = 8 - 0.08q \quad (17)$$

$$q_1^e = 68.18, \quad t_1^e = 2.55 \quad (18)$$

- Flow rises from 53.85 to 68.18 veh/tu: equilibrium shifts outward

Induced Demand: Equilibrium Shifts Outward

